Heart Disease Analysis

Phase 2: Requirement Analysis

Data Flow Diagram

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Overview

This diagram traces the end-to-end flow of data: from the raw Kaggle dataset, through database storage and SQL analysis, into Tableau visualization, and finally out to the end user via the published dashboard and embedded Flask web app.

End-to-End Data Flow

[Raw Dataset (Kaggle CSV, 4,500 rows, 18 columns)]

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[MySQL Database (heart_disease_db → table HD)]

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[SQL Analysis Queries (prevalence, demographics, risk factors, high-risk profiling)]

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[Tableau Desktop (Data Extract + Calculated Fields + Charts)]

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[Tableau Dashboard & Story (KPIs, Visualizations, Interactive Filters)]

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[Published to Tableau Public]

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[Embedded in Flask Web App (iframe)]

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[End User (Healthcare Stakeholder) — Insights & Decisions]

Stage Descriptions

Raw Dataset → Database: The CSV dataset is loaded into a MySQL table named HD within the heart_disease_db database.

Database → SQL Analysis: SQL queries aggregate and filter the data to produce metrics such as heart disease rate, age-group breakdowns, and lifestyle-risk correlations.

SQL Analysis → Tableau: Query results and the underlying table are connected to Tableau Desktop, where the live MySQL connection is converted to a static extract for portability.

Tableau → Dashboard & Story: Calculated fields and charts are assembled into an interactive dashboard and a six-part story.

Dashboard → Tableau Public: The finished workbook is published to Tableau Public, making it accessible via a shareable link.

Tableau Public → Flask App: The published view is embedded into a Flask web application using an iframe, giving it a custom UI wrapper.

Flask App → End User: The healthcare stakeholder views and interacts with the dashboard, drawing insights that inform action.